

Project: Automatic Grading

Functionalities Implemented:

- Generate expected outputs from instructor solutions.
- Grade each student's homework by comparing their outputs to expected outputs.
- Handle partial credit if outputs are partially correct.
- Update the main grade table with new grades.
- Clean up temporary files and directories after grading.

Bash Script with comments:

```
#!/bin/bash
# =====
# Automatic Grading Script
# =====

# -----
# Configuration
# -----
# Define file paths used in the grading process

ROSTER="instructor/roster"      # File containing list of student usernames
SOLUTION="instructor/solution"  # File containing correct commands (one per problem)
MAINTABLE="instructor/maintable" # Main grade table where scores will be stored

TMP_DIR="tmp_grading"          # Temporary directory used during grading
mkdir -p $TMP_DIR              # Create the temporary directory

# -----
# Count the number of lines in the solution file.
# Each line represents one problem/command.

NUM_PROBLEMS=$(wc -l < $SOLUTION)
# -----
# Step 1: Generate expected outputs
# -----
# This section runs the correct solution commands and stores
# their outputs so we can compare student answers against them.

echo "Generating expected outputs..."

# Loop through each problem number from 1 to NUM_PROBLEMS
for i in $(seq 1 $NUM_PROBLEMS); do
```

```

# Extract the i-th command from the solution file
# Each line of the solution file corresponds to one problem
CMD=$(sed -n "${i}p" $SOLUTION)

# Execute the command and store its output in a temporary file
# This file will contain the correct output for problem i
eval "$CMD" > "$TMP_DIR/expected_output_$i"
done

# -----
# Step 2: Grade students
# -----
# This section runs each student's commands, compares their outputs
# with the expected outputs, and calculates a grade.

echo "Grading students..."

# Create a temporary column that will store grades for HW1
GRADE_COLUMN="$TMP_DIR/grades_column"

# Add the header "HW1" for the new homework column
echo "HW1" > $GRADE_COLUMN

# Read each student username from the roster file
while read STUDENT; do

    # Each student has a directory named after their username
    STUDENT_DIR="$STUDENT"

    # Their homework file is named "hw" inside their folder
    STUDENT_HW="$STUDENT_DIR/hw"

    # Initialize total score for this student
    TOTAL=0

    # Loop through each problem
    for i in $(seq 1 $NUM_PROBLEMS); do

        # Extract the student's command for problem i
        STUDENT_CMD=$(sed -n "${i}p" $STUDENT_HW)

        # Run the student's command and store the output in a temp file
        eval "$STUDENT_CMD" > "$TMP_DIR/student_output"
    done
done

```

```

# Compare student output with the expected output
# diff prints differences; wc -l counts how many lines differ
DIFF_LINES=$(diff "$TMP_DIR/student_output" "$TMP_DIR/expected_output_$i" | wc -l)

# If there are no differences, give full credit
if [ $DIFF_LINES -eq 0 ]; then
    TOTAL=$(echo "$TOTAL + 1" | bc -l)

# If differences exist but are small (<10 lines), give partial credit
elif [ $DIFF_LINES -lt 10 ]; then
    TOTAL=$(echo "$TOTAL + 0.5" | bc -l)
fi
done

# Print the student's grade to the terminal
echo "$STUDENT got $TOTAL"

# Add the student's grade to the grade column file
echo "$TOTAL" >> $GRADE_COLUMN

# Continue reading students from the roster file
done < $ROSTER

# -----
# Step 3: Update main table
# -----
# This section adds the new homework grades to the main grade table.

# paste merges the existing table with the new grade column
paste $MAINTABLE $GRADE_COLUMN > "$TMP_DIR/temp_table"

# Replace the old main table with the updated version
mv "$TMP_DIR/temp_table" $MAINTABLE

# -----
# Step 4: Clean up
# -----
# Remove the temporary directory and files used during grading

rm -rf $TMP_DIR

```

```
# Final message confirming grading finished
```

```
echo "Grading complete! Main table updated at $MAINTABLE"
```

```
[jbui@mis01 ~/project] ./grade_script.sh
```

```
Generating expected outputs...
```

```
Grading students...
```

```
stu1 got 1.5
```

```
stu2 got 2.5
```

```
stu3 got 3
```

```
stu4 got 2.5
```

```
Grading complete! Main table updated at instructor/maintable
```

```
[jbui@mis01 ~/project] cat instructor/maintable
```

student	HW1
stu1	1.5
stu2	2.5
stu3	3
stu4	2.5